Related Work

This chapter reviews block-based tangible user interfaces (TUIs) that pursue goals similar to those of the system developed in this work, particularly the automatic detection of physical constructions assembled from building blocks. Physical building-block systems have become a promising form of TUI because their material properties and haptic feedback allow users to manipulate digital information through familiar physical actions. Existing approaches can broadly be divided into systems employing passive blocks and systems employing active blocks.

In this context, passive blocks do not perform local computation and do not require an individual power supply. Instead, their identity, position, orientation, or connectivity is determined by an external sensing system. Depending on the implementation, this detection may rely on computer vision, optical fibres, capacitive sensing, or RFID technology. Although passive RFID tags contain electronic components, they are classified as passive because they have no battery and obtain the required energy from an external reader.

Active blocks, by contrast, contain embedded electronic components, such as microcontrollers, sensors, or communication modules, and actively participate in the detection process. They may be powered by individual batteries or receive power through electrical connections between adjacent blocks. Active blocks can communicate directly with a host computer or exchange information with neighbouring blocks before forwarding the resulting topology data to a central system.

Passive block systems

Computer-vision-based detection

Computer-vision-based systems use one or more cameras to determine the identity, position, or orientation of physical blocks. DuploTrack employs such an approach to provide real-time assistance during the design and construction of LEGO Duplo models. An RGB-D camera captures both colour and depth information, from which the system reconstructs the arrangement of the blocks [citation].

A general limitation of camera-based detection is that the relevant parts of the construction must remain visible. Blocks may be temporarily occluded by other blocks or by the user's hands, potentially resulting in incomplete or incorrect detection. Moreover, the camera usually has to be positioned and calibrated according to the dimensions and lighting conditions of the interaction area.

Lumino addresses the detection of vertically stacked blocks using optical markers and light-conducting fibre bundles. Each block contains a fibre bundle that transfers its fiducial marker to the surface of an interactive table. An infrared camera integrated into the table captures the resulting composite marker image and uses it to reconstruct the stack [citation]. Because light is lost at the transitions between adjacent fibre bundles, however, the maximum stack height is limited to approximately five layers [citation].

Thus, both DuploTrack and Lumino rely on optical sensing, but they differ in how the visual information is made accessible to the camera. Whereas DuploTrack observes the construction externally, Lumino transfers information through the individual blocks to a camera beneath the interaction surface. The latter approach enables the detection of stacked blocks but requires specially constructed optical blocks and an instrumented table.

Capacitive detection

CapStones uses capacitive sensing to extend applications on conventional touchscreens with graspable physical objects. Conductive wires and contact pins inside the blocks transfer the capacitance of the user's body from the top of a stack to the touchscreen. This allows users to place

and move blocks on the display and combine them into vertical structures whose configurations can be interpreted by the application [citation].

The approach does not require batteries or active electronic components within the blocks. However, the blocks generally have to be touched by the user to remain detectable. Furthermore, the intrinsic capacitance of the conductive paths increases with the height of the stack. Once it exceeds the threshold supported by the touchscreen, reliable detection is no longer possible. Consequently, the reported stack height is limited to approximately three or four layers [citation].

ProtoBricks combines capacitive detection with colour sensing. Capacitive sensors integrated into the base detect the user's body through the conductive material of the blocks and thereby determine their spatial placement. A colour sensor additionally identifies the colour and, consequently, the type of the currently placed block [citation]. This combination provides more information than capacitive sensing alone. Nevertheless, the system cannot determine the precise rotation of individual blocks. In addition, the blocks have to be placed sequentially because simultaneous changes may be interpreted as a single interaction [citation].

Both systems demonstrate that capacitive sensing can enable passive blocks to interact with touch-sensitive surfaces. Their detection capabilities, however, depend on the conductive connection to the user and are constrained by the sensitivity and calibration of the underlying capacitive sensors.

RFID-based detection

Physical block constructions can also be detected using radio-frequency identification. In such systems, passive RFID tags receive their operating energy wirelessly from an external reader and return a stored identifier. This eliminates the need for batteries inside the blocks while allowing individual components to be distinguished.

RFIBricks uses modified ultra-high-frequency RFID tags to detect physical connections between blocks. Each tag is separated into an antenna and a chip component, which are connected through conductive neodymium magnets. The resulting circuit functions as a contact switch: the tag becomes readable only when the corresponding physical connection between two blocks is established [citation]. Four such switches are arranged on the relevant block surfaces. Different combinations of activated tag identifiers allow the system to determine the relative orientation of connected blocks. Because the number and density of required tags would otherwise become difficult to manage, the contacts are restricted to the upper and lower surfaces. The resulting constructions are therefore limited to one-dimensional stacks [citation].

RFIMatch is closely related to RFIBricks and similarly combines modified UHF RFID tags with magnetic elements. In contrast to RFIBricks, RFIMatch is designed to detect associations without requiring precisely aligned physical contacts. Its sensing mechanism is therefore less dependent on the exact orientation and alignment of the interacting objects [citation].

Both approaches operate in the UHF frequency range. The required UHF readers can be comparatively expensive, large, and power-intensive, particularly in relation to readers designed for shorter-range RFID or NFC applications [citation]. NFCStack therefore employs a lower-cost NFC reader based on the MFRC522. A multiplexed reader in the base station transmits its signal through antenna extensions integrated into the blocks. Depending on the block design, this signal is relayed through transmission lines or inductively coupled coils [citation]. Magnetic connectors or inductive coupling between neighbouring blocks continue the signal path towards the top of the stack.

This design allows NFCStack to identify the individual blocks and determine their position and orientation within a stack. Similar to RFIBricks, however, its physical connectivity is restricted to vertical stacking and therefore supports only one-dimensional constructions [citation].

RFIBricks, RFIMatch, and NFCStack all provide battery-free identification. Their scalability is nevertheless constrained by the properties of the radio-frequency communication channel. If many tags respond within the same reading area, collisions and limited communication bandwidth may reduce detection reliability. Furthermore, the achievable stack height is limited because the available signal strength decreases with increasing distance from the reader and with every additional coupling interface [citation].

Active block systems

Unlike passive systems, active block systems integrate electronic processing and communication components into the individual blocks. The blocks can therefore contribute directly to detecting the construction. The resulting topology information may be transferred through the physical chain of blocks, as in ActiveCube, or transmitted wirelessly by the individual blocks, as in eBlock and i-Cube.

Sequential wireless identification

eBlock is a tangible programming tool in which children arrange physical blocks into a sequence to specify the path of a virtual character. When activated, each block transmits its identifier wirelessly to a receiver connected to the host computer [citation]. Communication between neighbouring blocks is limited to the sequential activation process. A parent block uses an infrared signal to activate an adjacent child block. After sending its identifier to the computer, the child enables its own infrared transmitter and activates the next block in the sequence.

Because the connected blocks transmit their identifiers one after another, the host computer can infer their order from the arrival times of the messages and reconstruct the one-dimensional structure. This reduces the amount of direct communication required between neighbouring blocks. However, it also limits the representable topology to a sequential arrangement.

A related sequential principle is used by i-Cube, a tangible educational system designed to support cognitive development [citation]. Neighbouring blocks are detected locally using inductive sensors and a fire-and-latch protocol. Each square face contains four symmetrically arranged inductive receivers. The transmitter on the opposing face of a neighbouring block activates one of these receivers, allowing the system to determine both the connected faces and the relative rotation between the two blocks.

Each i-Cube then transmits the collected connection and orientation information to the host computer using its own radio module. Magnets arranged on the block faces maintain the physical connections. In contrast to eBlock, this design supports three-dimensional constructions and provides information about both connectivity and relative block orientation.

===Networked blocks with shared power

roBlocks also uses magnets, but employs them as both mechanical connectors and electrical contacts. Once connected, the blocks form an ad hoc communication network through which they exchange and broadcast their data [citation]. A dedicated communication block collects the network information and forwards it wirelessly to the host computer. The communication block also initiates topology discovery by querying the connected blocks and collecting their reports about local connections.

A separate power block supplies electrical energy to the network through the magnetic contacts. Consequently, the individual blocks contain embedded processing and communication components but do not require separate batteries. This distinguishes the system from active approaches in which every block has to be charged or equipped with a replaceable battery.

===Wired topology detection

ActiveCube similarly distributes power from a dedicated base cube. Adjacent blocks are connected through electrical terminals and additional mechanical hooks arranged on their faces. The blocks form a wired RS-485-based network and communicate using LON technology [citation]. Unlike systems in which every block communicates wirelessly with the host, ActiveCube transfers connection information through the physical network towards the base cube.

When a new block is attached, both participating cubes report the connection. The child cube transmits its cube identifier and the identifier of the connected face, while the parent cube reports its own identifier and the corresponding face. A dedicated input terminal on each face enables the child cube to determine which of its six faces has received power and has therefore been connected.

The host computer combines these messages to update a connection-status tree that represents the topology of the physical construction. The base cube acts as the bridge between the block network and the host. This architecture allows ActiveCube to reconstruct three-dimensional configurations in real time and has been applied to tasks such as three-dimensional modelling, model retrieval, flight simulation, and cognitive assessment [citation].

Integrated sensors and actuators

In addition to contributing to topology detection, active blocks can support local output and additional sensing. ActiveCube includes functional cubes containing components such as gyroscopic sensors, vibration actuators, and LED matrices. i-Cube provides illuminated faces and local audio output through vibration speakers. roBlocks includes specialised actuator blocks equipped with geared motors, piezoelectric speakers, or LEDs, whereas eBlock uses an LED on the upper surface of each block to communicate its current status [citations].

These components enable active systems to provide visual, auditory, or mechanical feedback directly at the point of interaction. Such local feedback is difficult to implement using entirely passive blocks because passive components generally cannot drive sensors, displays, motors, or other actuators without an external supply of energy.

Comparison of active and passive systems

Active and passive block systems present different advantages and limitations. Active blocks can detect local connections and communicate block identifiers directly, enabling detailed and potentially reliable reconstruction of complex three-dimensional structures. Their local energy supply also supports integrated sensors, lights, sound, vibration, and movement. This permits richer multimodal interaction and allows feedback to be presented directly through the physical construction.

These capabilities increase the hardware complexity of each block. Systems based on individual batteries require charging or periodic battery replacement. Even when power is distributed through the construction, the blocks require microcontrollers, communication circuitry, and electrical contacts. These components increase manufacturing costs and may reduce scalability when a large number of blocks is required.

Passive blocks have a simpler internal construction and generally require less block-level maintenance because they do not contain batteries. They are therefore potentially suitable for applications involving large numbers of physical components. However, the simplicity of the blocks shifts the technical complexity to the surrounding infrastructure, such as cameras, interactive tables, capacitive displays, or RFID readers. The complete system is therefore not necessarily simpler or less expensive.

The limitations of passive approaches depend strongly on the selected sensing principle. Camera-based systems are susceptible to occlusion and may require calibration. Optical and capacitive

systems are often restricted in their maximum stack height because of signal loss or increasing capacitance. RFID-based approaches can identify battery-free objects but may be affected by limited reading range, tag collisions, and attenuation through multiple layers. Furthermore, several passive systems support only vertical or otherwise one-dimensional constructions.

The selection between active and passive blocks therefore involves a trade-off between block-level complexity and the capabilities of the overall sensing system. Active blocks provide local processing, detailed topology information, and richer feedback, whereas passive blocks reduce the need for per-block power supplies and electronic maintenance. The suitability of either approach ultimately depends on the required construction space, number of blocks, recognition accuracy, interaction modalities, and acceptable hardware complexity.

Vor der Übernahme solltest du noch vier Angaben anhand der Originalpaper prüfen:

- Ob das System tatsächlich i-Cube, iBlock oder anders heißt.
- Ob der verwendete Begriff genau fire-and-latch lautet.
- ActiveCube verwendet deinen Notizen zufolge RS-485, nicht „RS-585“.
- Aussagen wie „reliable“, konkrete Stapelhöhen und technische Einschränkungen benötigen jeweils einen direkten Beleg aus dem betreffenden Paper.